



Figure 5: (Left) Graphical model representation of LDA. (Right) Graphical model representation of the variational distribution used to approximate the posterior in LDA.

5.1 Inference

The key inferential problem that we need to solve in order to use LDA is that of computing the posterior distribution of the hidden variables given a document:

$$p(\theta, \mathbf{z} | \mathbf{w}, \alpha, \beta) = \frac{p(\theta, \mathbf{z}, \mathbf{w} | \alpha, \beta)}{p(\mathbf{w} | \alpha, \beta)}.$$

Unfortunately, this distribution is intractable to compute in general. Indeed, to normalize the distribution we marginalize over the hidden variables and write Eq. (3) in terms of the model parameters:

$$p(\mathbf{w} | \alpha, \beta) = \frac{\Gamma(\sum_i \alpha_i)}{\prod_i \Gamma(\alpha_i)} \int \left(\prod_{i=1}^k \theta_i^{\alpha_i - 1} \right) \left(\prod_{n=1}^N \sum_{i=1}^k \prod_{j=1}^V (\theta_i \beta_{ij})^{w_n^j} \right) d\theta,$$

a function which is intractable due to the coupling between θ and β in the summation over latent topics (Dickey, 1983). Dickey shows that this function is an expectation under a particular extension to the Dirichlet distribution which can be represented with special hypergeometric functions. It has been used in a Bayesian context for censored discrete data to represent the posterior on θ which, in that setting, is a random parameter (Dickey et al., 1987).

Although the posterior distribution is intractable for exact inference, a wide variety of approximate inference algorithms can be considered for LDA, including Laplace approximation, variational approximation, and Markov chain Monte Carlo (Jordan, 1999). In this section we describe a simple convexity-based variational algorithm for inference in LDA, and discuss some of the alternatives in Section 8.

5.2 Variational inference

The basic idea of convexity-based variational inference is to make use of Jensen’s inequality to obtain an adjustable lower bound on the log likelihood (Jordan et al., 1999). Essentially, one considers a family of lower bounds, indexed by a set of *variational parameters*. The variational parameters are chosen by an optimization procedure that attempts to find the tightest possible lower bound.

A simple way to obtain a tractable family of lower bounds is to consider simple modifications of the original graphical model in which some of the edges and nodes are removed. Consider in particular the LDA model shown in Figure 5 (left). The problematic coupling between θ and β